

Technical Specification Sheet & Master Manufacturing Protocol: Advanced Bio-Spray Hair Oil

1. Executive Product Profile & Technical Specifications

Ethnobotanical hair and scalp treatments utilizing traditional Mediterranean plant lipids—specifically *Olea europaea* (olive) oil, *Ricinus communis* (castor) oil, *Prunus dulcis* (sweet almond) oil, and *Cocos nucifera* (coconut) oil—possess documented therapeutic value for managing alopecia, split hair ends, and seborrheic dermatitis. Quantitative ethnopharmacological field surveys in the West Bank, Palestine (Zaid et al.) document the widespread reliance on these natural vehicles: *Olea europaea* represents the primary traditional lipid vehicle ($UV = 1.38$, 369 citations), while *Ricinus communis* ($UV = 0.65$, 173 citations) exhibits high consensus factors for split hair endings ($F_{ic} = 0.95$) and hair loss ($F_{ic} = 0.93$). Furthermore, high informant consensus across key target indications—including hair endings ($F_{ic} = 0.95$), dandruff ($F_{ic} = 0.94$), hair loss ($F_{ic} = 0.93$), and hair damage ($F_{ic} = 0.93$)—validates the therapeutic necessity of these botanical agents. However, direct topical application of raw bulk oils suffers from severe organoleptic limitations, including excessive greasiness, poor spreadability, tacky residue, and low rinseability, leading to poor consumer compliance and secondary scalp irritation from harsh cleansing detergents. Formulating these ethnomedicinal botanical lipids into a sub-50 nm bio-based nanoemulsion resolves these compliance and mechanical delivery barriers. Unlike microemulsions, which are thermodynamically stable equilibrium systems requiring elevated surfactant concentrations ($>25 - 30\% \text{ w/w}$) that can disrupt the stratum corneum, nanoemulsions are *kinetically stable* non-equilibrium dispersions. Physical stability in nanoemulsion systems is governed by high kinetic energy barriers that prevent droplet collision, coalescence, and Ostwald ripening, achieved through high-energy processing (e.g., ultrasonic cavitation) rather than excessive surfactant loading. Encapsulating the lipophilic active matrix within sub-50 nm lipid nanocarriers yields an optically clear to translucent, sprayable mist that delivers rapid follicular and cuticle uptake, maximizes bioactive lipid bioavailability, and ensures complete water rinseability without greasy residues.

Core Engineering & Physicochemical Specifications

Specification Parameter, Technical Target / Release Criteria, Analytical Methodology
Mean Droplet Diameter (Z -Average), $< 50.0 \text{ nm}$, Dynamic Light Scattering (DLS) / Photon Correlation Spectroscopy
Polydispersity Index (PDI), < 0.200 (Monodisperse distribution), Cumulants Analysis via DLS at 25°C
Rheological Viscosity, $< 25.0 \text{ cP}$ at 25°C (Newtonian flow profile), "Rotational Viscometer (Spindle 1, 60 RPM)"
pH Specification Range, $5.00 - 5.50$ (Physiologically balanced), Calibrated Digital pH Probe with Temperature Compensation
Physical State & Appearance, "Single-phase, optically clear to translucent", Photometric Transmittance ($T_{600 \text{ nm}} \geq 85\%$) / Visual Inspection

Delivery Mechanism, Micro-spray aerosol or mechanical pump misting, "Mechanical Actuator ($D_{v,50} = 20 - 50 \mu\text{m}$ aerosol droplets)"

Analytical Evaluation & Functional Interdependence

The functional relationship between maintaining a bulk fluid viscosity below 25.0 cP , a mean droplet diameter below 50.0 nm , and a narrow Polydispersity Index ($\text{PDI} < 0.200$) is critical to physical kinetic stability and spray performance:

- **Kinetic Stability & Creaming Inhibition:** Hydrodynamic velocity of particle sedimentation or creaming is governed by Stokes' Law: $v = \frac{2}{9} \frac{g r^2 (\rho_p - \rho_f)}{\eta}$ where r is the droplet radius, g is gravitational acceleration, $\rho_p - \rho_f$ is the density differential between the dispersed oil phase and continuous aqueous phase, and η is continuous phase viscosity. Downsizing the dispersed lipid droplets to $r < 25 \text{ nm}$ (Z-average $< 50.0 \text{ nm}$) reduces the velocity of gravitational separation by several orders of magnitude. At this scale, Brownian motion overrides gravitational forces, suppressing creaming and sedimentation. Maintaining $\text{PDI} < 0.200$ prevents Ostwald ripening, wherein small droplets dissolve and diffuse into larger droplets due to Laplace pressure differentials ($\Delta P = 2\gamma/r$).
- **Low-Shear Rheology & Actuator Atomization:** A bulk viscosity target below 25.0 cP guarantees near-Newtonian rheological behavior during low-shear storage and high-shear actuation. This low viscosity allows the fluid to pass through the orifice of a mechanical mist actuator without shearing instabilities or nozzle clogging, achieving consistent atomization into fine aerosol plumes.
- **Follicular Absorption & Interfacial Area:** Sub-50 nm lipid nanocarriers drastically expand the surface-area-to-volume ratio ($A/V = 3/r$), maximizing contact efficiency with the hydrophobic hair shaft and follicular infundibulum. Rapid transdermal uptake eliminates surface oil accumulation, ensuring a light, non-tacky feel and easy removal during rinsing. Proceed to Phase Architecture setup to balance required HLB prior to thermal shearing.

2. Master Formulation Ratios & Ingredient Architecture

The master formulation is engineered as a three-phase system consisting of a **Lipophilic Carrier Complex (Phase A)**, a **Surfactant & Nanocarrier Stabilizer System (Phase B)**, and an **Aqueous & Botanical Extract Phase (Phase C)**. Ethnobotanical oils are selected based on their specific fatty acid profiles and quantitative therapeutic indices: *Olea europaea* (rich in oleic acid, $C_{18:1}$), *Ricinus communis* (dominated by ricinoleic acid, $C_{18:1}\text{-OH}$), *Prunus dulcis* (oleic/linoleic matrix), *Cocos nucifera* (lauric acid, $C_{12:0}$), and *Simmondsia chinensis* (liquid wax esters). To optimize the lipophilic core for atomization, Palm Kernel Oil Esters (PKOEs)—specifically synthesized low-viscosity fatty acid methyl/ethyl esters rich in caprylic ($C_{8:0}$), capric ($C_{10:0}$), and lauric ($C_{12:0}$) chains—are integrated. PKOEs act as a specialized ester co-solvent that reduces bulk lipid phase viscosity, lowers interfacial tension, and enhances spray atomization. Phase C incorporates aqueous botanical actives including *Aloe vera* gel extract, *Rosmarinus officinalis* hydro-distillate, *Lavandula coronopifolia* extract, and a decolorized, purified active fraction of

Lawsonia inermis ($UV = 0.84$, 223 citations, $FL = 57.8\%$). Using a decolorized fraction of *Lawsonia inermis* isolates bioactive lawsone-complex derivatives and polyphenols to preserve antifungal, antimicrobial, and scalp conditioning efficacy while removing naphthoquinone pigment fractions that would otherwise cause permanent henna staining or clouding in a sprayable liquid mist. To stabilize Phase A within Phase C, Phase B utilizes a synergistic amphiphilic surfactant blend. Polysorbate 80 (Non-ionic Surfactant, Hydrophilic-Lipophilic Balance $HLB = 15.0$) is paired with *Polyglyceryl-10 Laurate* (Co-Surfactant, $HLB \approx 16.0$) and *1,3-Propanediol* (Co-Solvent / Humectant). The required HLB (HLB_{req}) of the combined Phase A lipid complex is calculated at $12.5 - 14.0$. The primary surfactant and co-surfactant are combined in explicit ratios to match this HLB_{req} , forming a rigid interfacial monomolecular film around the nano-droplets that prevents coalescence.

Master Formulation Ratios (Formulation A vs. Formulation B)

Phase, Ingredient / INCI / Chemical Name, Function / Specification, Standard Nanoemulsion (Formulation A) % w/w, High-Load Bio-Spray Oil (Formulation B) % w/w		
Phase A, <i>Olea europaea</i> (Olive) Fruit Oil, Lipophilic Vehicle ($UV = 1.38$),	8.00,	12.00
Phase A, <i>Ricinus communis</i> (Castor) Seed Oil, Conditioning Lipid ($UV = 0.65$),	4.00,	6.00
Phase A, <i>Prunus dulcis</i> (Sweet Almond) Oil, Emollient ($UV = 0.28$),	3.00,	4.00
Phase A, <i>Cocos nucifera</i> (Coconut) Oil, Penetrating Fatty Acid ($UV = 0.31$),	2.00,	3.00
Phase A, <i>Simmondsia chinensis</i> (Jojoba) Seed Oil, Cuticle Sealer ($UV = 0.02$),	1.00,	2.00
Phase A, Palm Kernel Oil Esters (PKOEs), Low-Viscosity Ester Co-Solvent,	2.00,	3.00
Phase B, Polysorbate 80, Primary Non-ionic Surfactant ($HLB = 15.0$),	10.00,	14.00
Phase B, Polyglyceryl-10 Laurate, Co-Surfactant ($HLB \approx 16.0$),	3.00,	4.00
Phase B, "1,3-Propanediol", Co-Solvent / Humectant / Coupling Agent,	2.00,	2.00
Phase C, <i>Aloe vera</i> Gel Extract, Aqueous Soothing Active ($UV = 0.32$),	5.00,	4.00
Phase C, <i>Rosmarinus officinalis</i> Hydro-distillate, Scalp Toning Active ($UV = 0.03$),	4.00,	3.00
Phase C, <i>Lavandula coronopifolia</i> Extract, Aromatic / Calming Active ($UV = 0.01$),	2.00,	1.00
Phase C, Decolorized <i>Lawsonia inermis</i> Extract Fraction, Purified Antifungal Active ($UV = 0.84$),	1.00,	2.00
Phase C, Purified Water, Aqueous Continuous Base,	53.00,	40.00
TOTAL,,,	100.00,	100.00

Comparative Formulation Analysis & Surfactant Loading Metrics

- Formulation A (Standard Botanical Nanoemulsion):** Contains 20.00% total lipid phase (Phase A) and 15.00% surfactant/co-solvent phase (Phase B), yielding an oil-to-surfactant ratio (S_{or}) of 1.33. With an aqueous load of 65.00% , Formulation A maintains a low bulk viscosity (< 15.0 cP), enabling mist dispersion for daily preventative maintenance against hair loss, light scalp conditioning, and anti-dandruff control.
- Formulation B (Enriched / High-Load Bio-Spray Oil):** Expands total lipid payload to 30.00% (Phase A) stabilized by 20.00% Phase B ($S_{or} = 1.50$). The elevated lipid payload increases viscous drag and interfacial resistance during emulsification. Consequently, achieving a Δ -average < 50.0 nm in


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| Yields: Coarse Macro-Emulsion (Droplet Size: 1.0 - 5.0 μm)
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|                               ULTRASONIC CAVITATION NANO-SIZING
|
| 1.5 kW Industrial Ultrasonic Probe @ 20 kHz | Amplitude: 80%
|
| Inline Flow Cell | Backpressure: 2.0 - 3.0 bar | Flow Rate: 0.5 -
1.0 L/min      |
| Energy Density: 150 - 220 J/mL | Continuous Chilled Cooling (Temp
Output < 30°C) |
| Yields: Stable Sub-50 nm Translucent Nanoemulsion
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Phase-by-Phase Preparation Protocols

- 1. Phase A (Lipid Phase Preparation):** Charge a jacketed stainless steel batch vessel with *Olea europaea* oil (\$8.00\%\text{ w/w}\$ Form A / \$12.00\%\text{ w/w}\$ Form B), *Ricinus communis* oil (\$4.00\% / 6.00\%\text{ w/w}\$), *Prunus dulcis* oil (\$3.00\% / 4.00\%\text{ w/w}\$), *Cocos nucifera* oil (\$2.00\% / 3.00\%\text{ w/w}\$), *Simmondsia chinensis* oil (\$1.00\% / 2.00\%\text{ w/w}\$), and Palm Kernel Oil Esters (\$2.00\% / 3.00\%\text{ w/w}\$). Initiate low-shear variable-speed propeller agitation at \$200 - 300\text{ RPM}\$ while heating the vessel jacket to \$40^{\circ}\text{C} - 45^{\circ}\text{C}\$. Maintain thermal input until all lipids are liquefied and uniform. Do not exceed \$45^{\circ}\text{C}\$ to prevent thermal degradation of phenolic antioxidants and unsaturated fatty acid double bonds.
- 2. Phase B & C Integration Protocol:** In a secondary vessel, blend purified water with *Aloe vera* gel extract, *Rosmarinus officinalis* hydro-distillate, *Lavandula coronopifolia* extract, and the decolorized *Lawsonia inermis* fraction at ambient temperature (\$20^{\circ}\text{C} - 25^{\circ}\text{C}\$). In a separate feed vessel, pre-mix Phase B components (Polysorbate 80, Polyglyceryl-10 Laurate, and 1,3-Propanediol) and add directly into Phase A at \$40^{\circ}\text{C}\$. Meter the combined Phase A/B oil-surfactant blend into Phase C under continuous mechanical stirring at \$500\text{ RPM}\$ for \$10\text{ minutes}\$ to form an initial dispersion.

High-Shear Pre-Mixing Protocol

Transfer the crude multi-phase blend to a high-shear rotor-stator homogenizer (e.g., Silverson Inline Homogenizer) to generate a coarse macro-emulsion:

- **Rotational Velocity & Shear Rate:** Drive rotor speed to $6,000 - 8,000 \text{ RPM}$, generating a localized shear rate of approximately $\dot{\gamma} \approx 10,000 \text{ s}^{-1}$ within the rotor-stator gap.
- **Temperature Control:** Recirculate cooling water through the jacketed vessel to retain batch processing temperature strictly between 25°C and 35°C .
- **Processing Duration:** Maintain continuous high-shear processing for 15 minutes .
- **Target Intermediate Criteria:** Yields a uniform, opaque coarse macro-emulsion with droplet diameters distributed between $1.0 \mu\text{m}$ and $5.0 \mu\text{m}$.

High-Energy Ultrasonic Cavitation Protocol

Pump the coarse macro-emulsion continuously through an inline high-energy ultrasonic flow-cell system equipped with a 1.5 kW industrial titanium probe horn operating at a fixed frequency of 20 kHz :

- **Acoustic Power & Amplitude:** Set generator output to 80% operational amplitude (1.2 kW effective acoustic power output).
- **Flow Rate & Fluid Hydraulics:** Meter product throughput at $0.5 - 1.0 \text{ L/min}$ through the flow cell while maintaining an inline system backpressure of $2.0 - 3.0 \text{ bar}$. Cell backpressure intensifies the symmetrical collapse of cavitation bubbles, amplifying acoustic micro-jets and transient shear forces necessary to disrupt macro-droplets below the 50 nm threshold.
- **Volumetric Energy Density Input:** Deliver a minimum acoustic energy density of 150 J/mL for Formulation A and 220 J/mL for Formulation B.
- **Thermal Management:** Recirculate glycol coolant ($2^{\circ}\text{C} - 5^{\circ}\text{C}$) through the flow-cell cooling jacket. Product discharge temperature must be maintained strictly below 30°C to preserve active botanical polyphenols and volatile terpenes.

Master Processing Sequence & Quality Checkpoints

Step #, Process Phase, Operational Equipment, Operational Parameters, Critical Quality Checkpoints

01, Phase A Lipid Homogenization, Jacketed Tank / Propeller Agitator, 250 RPM ; $40^{\circ}\text{C} - 45^{\circ}\text{C}$; 15 min , "Homogeneous, clear liquid phase; zero undissolved fats"

02, Phase B/C Solubilization, Stainless Tank / Paddle Mixer, 500 RPM ; $20^{\circ}\text{C} - 25^{\circ}\text{C}$; 10 min , Complete aqueous solubilization; clear particle-free phase

03, High-Shear Pre-Emulsification, High-Shear Rotor-Stator Mixer, $7,000 \text{ RPM}$ ($\dot{\gamma} \approx 10,000 \text{ s}^{-1}$); $25^{\circ}\text{C} - 35^{\circ}\text{C}$; 15 min , Macro-emulsion uniformity; mean droplet size $1 - 5 \mu\text{m}$

04, Ultrasonic Cavitation, 1.5 kW Flow-Cell Ultrasonic System, 20 kHz ; 80% Amplitude; $0.5 - 1.0 \text{ L/min}$; Backpressure $2 - 3 \text{ bar}$; Temp $< 30^{\circ}\text{C}$, Optical clarity transition; ZS -average $< 50.0 \text{ nm}$; $\text{PDI} < 0.200$

05, Micro-Filtration & Transfer, Inline $0.22\ \mu\text{m}$ Cartridge Filter, Closed sanitary loop; ambient pressure, Particulate-free; viscosity $< 25.0\ \text{cP}$; pass pH specification
Proceed to analytical release testing and stability certification to validate batch compliance against nanocarrier release criteria.

4. Quality Control, Release Specifications & Stability Standards

To certify commercial release and regulatory compliance, every production batch of Advanced Bio-Spray Hair Oil must undergo analytical testing against defined physical, chemical, and organoleptic quality limits.

Master Quality Control Release Criteria

Test Parameter, Analytical Testing Protocol, Pass / Fail Release Limits

Mean Droplet Diameter (Z -Average), Dynamic Light Scattering (DLS) at 25°C , $< 50.0\ \text{nm}$

Polydispersity Index (PDI), DLS Cumulants Analysis, < 0.200

Rheological Viscosity, "Rotational Viscometer (Spindle 1, 60 RPM at 25°C)", $< 25.0\ \text{cP}$

pH Specification, Calibrated Digital pH Probe at 25°C , $5.00 - 5.50$

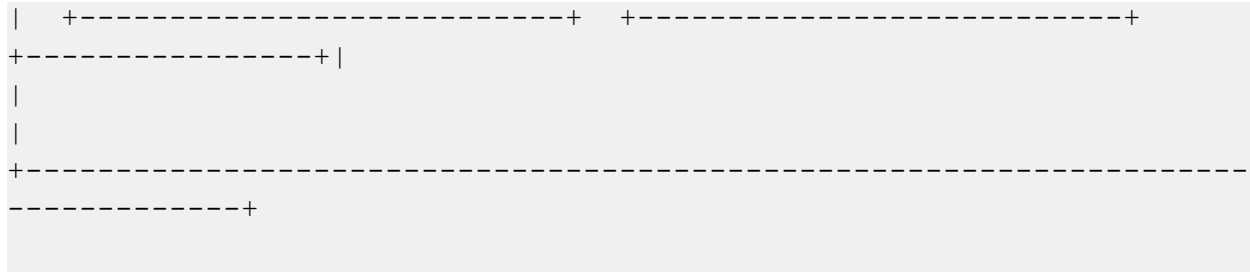
Centrifugal Phase Stability, "Analytical Centrifugation at $4,000\ \text{RPM}$ for $30\ \text{min}$ (25°C)", "Zero creaming, phase separation, or sediment formation"

Peroxide Value (PV), Iodometric Titration (ISO 3960), $< 5.0\ \text{meq O}_2/\text{kg}$ lipid phase

Acid Value (AV), Titrimetric Potassium Hydroxide Assay (ISO 660), $< 1.0\ \text{mg KOH/g}$ lipid phase

Organoleptic & Appearance, Photometric Transmittance ($T_{600\ \text{nm}}$) & Sensory Evaluation, Optically clear to translucent ($T_{600\ \text{nm}}$ $\geq 85\%$); characteristic herbal scent; non-tacky feel

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ACCELERATED THERMAL STRESS THERMAL FREEZE-THAW			
SPRAY MIST			
$40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / 75% RH 5 Cycles (-20°C to $+40^{\circ}\text{C}$)			
CHARACTERIZATION			
$\text{PV} < 5.0\ \text{meq O}_2/\text{kg}$ $\Delta Z\text{-Avg} < 5.0\ \text{nm}$			
Plume: $30^{\circ} - 45^{\circ}$			
$\text{AV} < 1.0\ \text{mg KOH/g}$ $\text{PDI} < 0.200$			
Dv, 50: 20-50 μm			



Stability Assessment Protocols & Shelf-Life Validation

- Accelerated Thermal Stress & Lipid Oxidation Protocols:** Batch samples are stored in sealed, headspace-purged amber glass containers at $40^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and $75\% \pm 5\%$ Relative Humidity (RH) for 6 months in accordance with ICH guidelines. Quantitative oxidative markers are evaluated monthly to monitor hydrolytic and radical degradation of unsaturated fatty acids from castor and olive oils:
 - Peroxide Value Limit:* Must remain $\text{PV} < 5.0 \text{ meq O}_2/\text{kg}$.
 - Acid Value Limit:* Must remain $\text{AV} < 1.0 \text{ mg KOH/g}$.
 - Nanocarrier Size Stability:* Mean droplet size must not shift by more than $\Delta Z_{\text{-Average}} > 5.0 \text{ nm}$ from baseline values.
- Thermal Freeze-Thaw Stress Cycling:** Samples undergo 5 consecutive freeze-thaw cycles consisting of 24 hours at $-20^{\circ}\text{C} \pm 2^{\circ}\text{C}$ followed by 24 hours at $+40^{\circ}\text{C} \pm 2^{\circ}\text{C}$. Post-cycling analysis via DLS must confirm $Z_{\text{-average}} < 50.0 \text{ nm}$ and $\text{PDI} < 0.200$. Any droplet growth, phase separation, or PDI expansion beyond 0.200 indicates Ostwald ripening or film collapse and constitutes a lot failure.
- Actuator Delivery & Spray Pattern Characterization:** Photostability and mechanical compatibility testing are executed by packaging the nanoemulsion into mechanical mist pump bottles fitted with standard spray actuators. Samples are exposed to simulated ambient lighting ($1.2 \text{ million Lux hours}$) and tested for mechanical spray performance:
 - Spray Pattern Angle:* Must maintain a fully developed plume geometry with an angle between 30° and 45° .
 - Actuated Aerosol Particle Size:* Laser diffraction analysis (e.g., Malvern Spraytec) of the actuated aerosol mist must yield a volumetric median droplet diameter of $D_{v,50} = 20 - 50 \text{ }\mu\text{m}$.
 - Nozzle Clogging Rate:* Must exhibit zero orifice buildup, jetting, or actuation failure across 1,000 continuous pump actuations. Strict compliance with these master formulation ratios, mechanical homogenization protocols, and analytical quality standards certifies that each batch of Advanced Bio-Spray Hair Oil meets commercial cosmeceutical and chemical engineering requirements for release.